Midterm Problem 1-2

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Problem 1

The following code simulates and plots 5000 trajectories of Geometric Brownian motion with starting value $X_0 = 100$, growth rate$\mu = 0.10$, and volatility $\sigma = 0.21$.

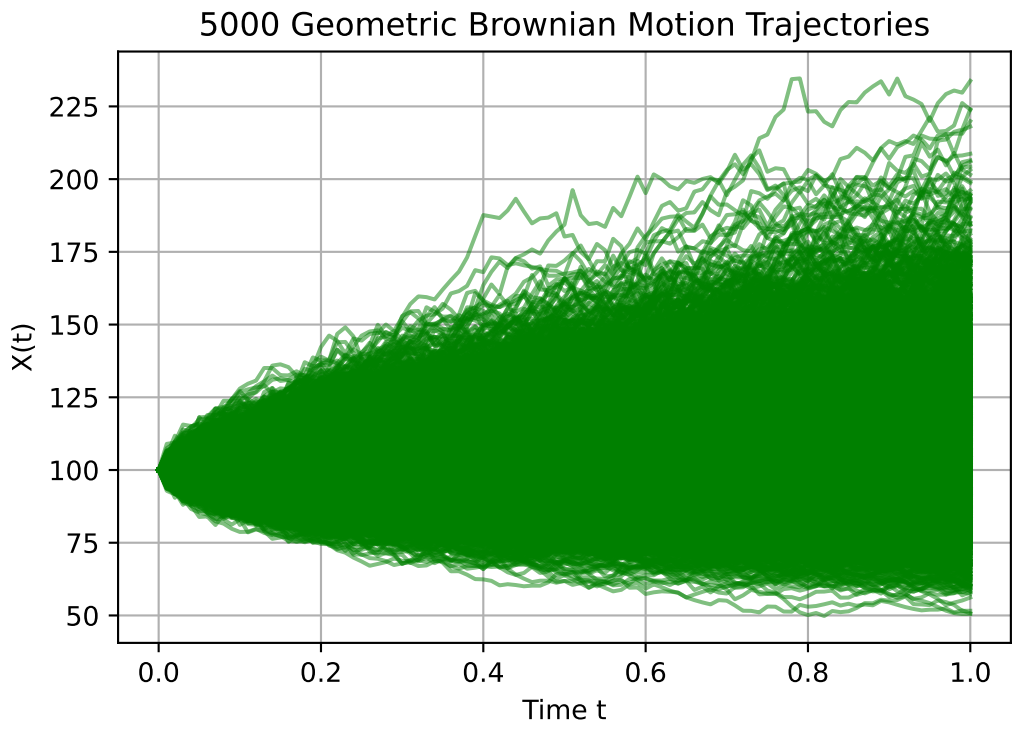
```
import numpy as np
import matplotlib.pyplot as plt

n_paths = 5000
n_steps = 100
T = 1.0
dt = T / n_steps
Sqrtdt=np.sqrt(dt)
rng = np.random.default_rng(17) #initialize rand numb generator
X = np.zeros((n_paths, n_steps+1)) #Allocate a zero array of trajectories

#GBM setup
X0 = 100
mu = 0.10
sigma = 0.21

X[:, 0] = X0 # initial starting point to X0
for i in range(n_paths):
    for j in range(1,n_steps+1):
      z = rng.normal(loc=0, scale=1)
      # GBM: X_(t + dt) = X_t * exp{(mu - 1/2 * sigma^2)dt + sigma * sqrt(dt)z}
      X[i, j] = X[i, j-1] * np.exp((mu - 0.5 * sigma**2) * dt + sigma * Sqrtdt * z)

t = np.linspace(0, T, n_steps + 1)
# Plotting
plt.figure(figsize=(6, 4))
plt.plot(t, X.T, alpha=0.5, color='green')
plt.title('5000 Geometric Brownian Motion Trajectories')
plt.xlabel('Time t')
plt.ylabel('X(t)')
plt.grid(True)
plt.show()
```



Problem 2

The code plots the Black–Scholes call price with dividend yield $q = 3.2\%$ and strike $K = 11.50$, alongside its expiration payoff.

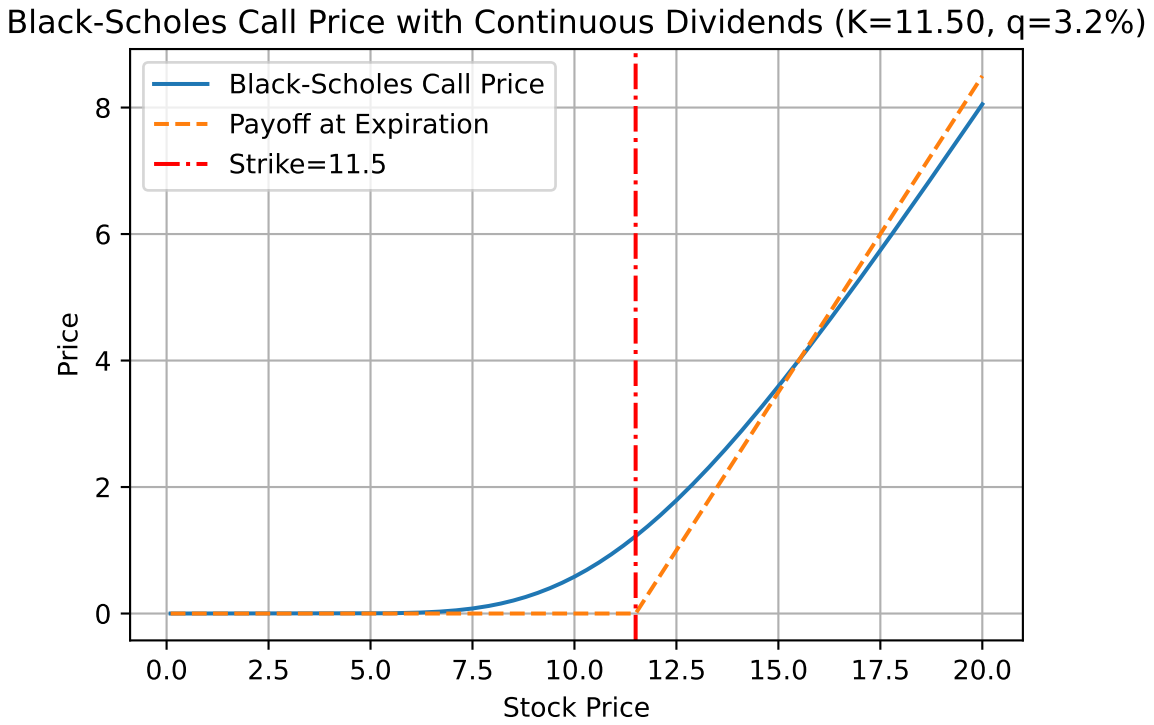
```
import math
import statistics
import matplotlib.pyplot as plt
import numpy as np

def BlackScholesStocks(callput, S, K, r, sigma, T, q=0.0):
    # d1 = (ln(s/k) + (r-q + 1/2 * sigma^2)T)/(sigma*sqrt(T))
    d1 = (math.log(S / K) + (r - q + 0.5 * sigma**2) * T) / (sigma * math.sqrt(T))
    d2 = d1 - sigma * math.sqrt(T)
    if callput == 'c':
        N1 = statistics.NormalDist().cdf(d1)
        N2 = statistics.NormalDist().cdf(d2)
        price = S * math.exp(-q * T) * N1 - K * math.exp(-r * T) * N2
    elif callput == 'p':
        N1 = statistics.NormalDist().cdf(-d1)
        N2 = statistics.NormalDist().cdf(-d2)
        price = K * math.exp(-r * T) * N2 - S * N1 * math.exp(-q * T)
    else:
        raise ValueError("callput argument must be 'c' for call or 'p' for put")
    return price
```

```
dx = 0.1  # step to evaluate option for graphing
maxX = 20.0  # max value of Stock price on graph
X = np.arange(dx, maxX + dx, dx)

Strike = 11.5 #update K=11.5
q = 0.032 # dividend = 3.2%
Rate = 0.01
Time = 1.0
Volatility = 0.3
Y = np.maximum(X - Strike, 0)  # array of Call payoff at expiration
Z = np.zeros_like(X)
for i, x_val in enumerate(X):
    Z[i] = BlackScholesStocks("c", x_val, Strike, Rate, Volatility, Time, q=q)

plt.figure(figsize=(6, 4))
plt.title("Black-Scholes Call Price with Continuous Dividends (K=11.50, q=3.2%)")
plt.plot(X, Z, label="Black-Scholes Call Price")
plt.plot(X, Y, label="Payoff at Expiration", linestyle="--")
plt.xlabel("Stock Price")
plt.ylabel("Price")
plt.axvline(x=Strike, linestyle='-.', label=f'Strike={Strike}', color='red')
plt.legend()
plt.grid(True)
plt.show()
```



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Q3. Proof. Let C and c be the prices of an American and European calls with the same strike price K and maturity T.

(a) Since an American option can provide the same exercise intrinsic value as the European option, plus additional early exercise opportunity,

$$C \geq c$$

For a no-divident paying stock, the no-arbitrage lower bound for European call is:

$$c \geq S_0 - Ke^{-rT}$$

Therefore, since $r>0$, $T>0$.

$$C \geq c \geq S_0 - Ke^{-rT} > S_0 - K$$

Since $C > S_0 - K$, we proved the American call worth more than early exercise value $S_0 - K$ so should not be exercise early for an American call on non-divident paying stock.

If the American call is held until maturity, both options have identical payoffs. Therefore, $C > c$ creates an arbitrage opportunity and break no arbitrage condition. Thus,

$$C = c$$

(b) Since the above we proved that

$$C > S_0 - K.$$

We proved American call on non-dividend pay stock worth more than early exercise, it can never be optimum to exercise early.

Q4 (a) When dividends exists, sometimes it is optimal to exercise an American call option prior to an ex-divident date to receive the divident.

For a divident paying stock, the European call may fall below its intrinsic value $S_T - K$. Thus selling the european call may not cover the cost of an early exercise of an American call. The arbitrage argument in Q3 no longer applies.

(b) Choose $x = 1$, $K = 100$, $r = 10\%$, $T - t = 1$ year, $\delta = 10\%$

Early exercise on a put, you will get $K - x = 99$ today.

For european put, since you can only exercise at maturity and $S_T \geq 0$, the pay off will be smaller than $\max(K - S_T, 0)e^{-r(T-t)} < K \cdot e^{-r(T-t)} = 100 \cdot e^{-0.1} \approx 90.48 < 99$

Therefore we prove

$$P_{American} > P_{european}.$$

Q5 (a) Apply the BS model (no divident)

$$C = S_0 N(d_1) - Ke^{-rT} N(d_2)$$

$$d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\delta^2)T}{\delta\sqrt{T}}$$

$$d_2 = d_1 - \delta\sqrt{T}$$

$S_0 = 100$, $K = 101$, $r = 0.041$, $\delta = 0.24$, $T = \frac{3}{12} = 0.25$

$$d_1 = \frac{\ln(100/101) + (0.041 + \frac{1}{2}0.24^2) \cdot 0.25}{0.24 \cdot \sqrt{0.25}} \approx 0.0625$$

$d_2 = 0.0625 - 0.24 \cdot \sqrt{0.25} \approx -0.05750$

$N(d_1) \approx 0.5249$ $N(d_2) \approx 0.4771$

$C = 100 \cdot 0.5249 - 101 e^{-0.041 \cdot 0.25} \cdot 0.4771 \approx 4.799$

Thus, the european call is approximately 4.80 dollars

delta

(b) $\Delta = N(d_1) \approx 0.5249$

gamma $\Gamma = \dfrac{\phi(d_1)}{S_0 \sigma \sqrt{T}} \approx 0.0332$

Rho: $\rho = K T \cdot e^{-rT} N(d_2) \approx 11.9232$ $\quad \text{Rho}_{1\%} = \dfrac{\rho}{100} = 0.119232$

Vega $= S_0 \sqrt{T}\, \phi(d_1) \approx 19.9082$ $\quad \text{Vega}_{1\%} = \dfrac{\text{Vega}}{100} \approx 0.199082$

Approximate using option calculator with days to expiration = 91

$C = \$4.79$

$\Delta = 0.5249$

$\Gamma = 0.0332$

$\text{Rho}_{1\%} = 0.1189$

$\text{Vega}_{1\%} = 0.1989$

(c) $\Theta = -\dfrac{S_0 \sigma \phi(d_1)}{2\sqrt{T}} - rKe^{-rT} N(d_2) \approx -11.5113$

apply $P = 4.80$, $\Delta = 0.5249$, $\Gamma = 0.0332$ from (a) and (b)

$LHS = \Theta + rS\Delta + \frac{1}{2}\sigma^2 S^2 \Gamma = -11.5113 + 0.041 \cdot 100 \cdot 0.5249 + \frac{1}{2}(0.24)^2 \cdot (100)^2 \cdot 0.0332$

$= 0.20239$

$RHS = rP = 0.041 \cdot 4.80 = 0.1968$

$LHS \approx RHS$, the small differences are from rounding errors.

Q6 (a) Factors affecting European and American option prices

1. Current stock price : S_0
2. Strike price : K
3. Time to expiry : T
4. Stock price volatility σ
5. Risk-free rate : r
6. Dividend : d

(b) ↑ means increase, ↓ means decrease when the factors remain the same; the same; ? means uncertain

	European Call	European Put	American Call	American Put
Stock price S_0 ↑	↑	↓	↑	↓
Strike price K ↑	↓	↑	↓	↑
Time to maturity T ↑	?	?	↑	↑
Volatility σ ↑	↑	↑	↑	↑
Risk-free rate r ↑	↑	↓	↑	↓
Divident d ↑	↓	↑	↓	↑

Q7 Yes. We can arbitrage by constructing a buttery fly spread:

(1) Buy 1 European call with $K_1 = 30$ for \$10
(2) Sell 2 European call with $K_2 = 35$ for \$8 x 2 = \$16
(3) Buy 1 European call with $K = 40$ for \$5

Initially, we receive \$16 - \$10 - \$5 = \$1. With following payoff structure:

	Long call $K_1 = 30$	Short 2 Call $K_2 = 35$	Long call $K_3 = 40$	Total
$S_T \leq 30$	0	0	0	0
$30 < S_T \leq 35$	$S_T - 30$	0	0	$S_T - 30$
$35 < S_T \leq 40$	$S_T - 30$	$-2(S_T - 35)$	0	$40 - S_T$
$S_T \geq 40$	$S_T - 30$	$-2(S_T - 35)$	$S_T - 40$	0

(S_T is the price at maturity)

Since the payoff at maturity ≥ 0 and the initial payoff is \$1, there is arbitrage opportunity.

Q8 Since $dX_t = \mu X_t dt + \sigma X_t dW_t$, $X_0 = 50$, $T = 1$

$X_0 = 50$, $T = 1$

$\mu = 0.11$, $\sigma = 0.22$

(a) pdf.

For geometric brownian motion:

$$X_T = X_0 \exp\left[\left(\mu - \frac{\sigma^2}{2}\right)T + \sigma W_T\right]$$

Since $W_T \sim N(0, T) = N(0, 1)$ given $T = 1$

$$\ln X_T \sim N\left(\ln X_0 + \left(\mu - \frac{\sigma^2}{2}\right)T, \sigma^2 T\right)$$

$$= N\left(\ln 50 + \left(0.11 - \frac{0.22^2}{2}\right), 0.22^2\right)$$

$$= N(3.997823, 0.0484)$$

Therefore $X_T = X_1$ follows a lognormal distribution with probability density function

$$f_{X_T} = \frac{1}{x\sqrt{2\pi\sigma^2 T}} \exp\left\{-\frac{\left(\ln(x/X_0) - (\mu - \sigma^2/2)T\right)^2}{2\sigma^2 T}\right\}, \quad x > 0$$

Apply $T = 1$, $\sigma = 0.22$, $\mu = 11$, $X_0 = 50$,

$$f_{X_1} = \frac{1}{0.22x\sqrt{2\pi}} \exp\left\{-\frac{(\ln(x/50) - 0.0858)^2}{2(0.22)^2}\right\}, \quad x > 0$$

(2) mean and std.

Since $E[X_T] = X_0 e^{\mu T}$, $SD(X_T) = X_0 e^{\mu T}\sqrt{e^{\sigma^2 T} - 1}$

$$E[X_1] = 50 \cdot e^{0.11} = 55.8139$$

$$SD(X_1) = 50 e^{0.11}\sqrt{e^{0.22^2} - 1}$$

$$= 12.4291$$